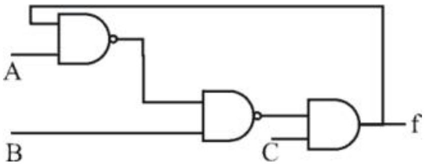
1. **Consider the circuit shown below. In a certain steady state, Y is at logical 'I'. What are possible values of A, B, C?**



(a) A = 0, B = 0, C = 1
(b) A = 0, B = C = 1
(c) A = 1, B = C = 0
(d) A = B = 1, C = 1

Ans. (a) : $f = \overline{\overline{Af}.B}.C = \left(Af + \overline{B}.\right)C$

When the output is logic 1 the output equation is 1=(Af+ $\overline{B}$).C To justify this equation 'C' must be always '1'

So, C=1, A=1,B=0

Or A=1, B=1

Or A=0, B=0

2. **The worst case time complexity of AVL tree is better in comparison to binary search tree for**

(a) Search and Insert Operations
(b) Search and Delete Operations
(c) Insert and Delete Operations
(d) Search, Insert and Delete Operations

Ans. (d) : Because search is O(log n) since AVL tree are always balanced, insertion & detection are also 0 (log n)

where as in case of BST is O(n)

3. **The GSM network is divided into the following**

(a) SS, BSS, OSS (b) BSS, BSC, MSC
(c) CELL, BSC, OSS (d) SS, CELL, MSC

Ans. (a) : SS, BSS, OSS

Because, GSM describe the protocols for second generation (ZG) digital cellular network used by mobile phones, GSM supports voice call & data transfer speed of up to 9.6 kbps. It is based on the time division multiple access (TDMA) system. Currently GSM Network operate on the 850 Mhz, 900 Mhz, 1800 Mhz, 1900 Mhz frequency band.

4. **The power set of the set $\{\varphi\}$ is**

(a) $\{\phi\}$ (b) $\{\phi, \{\phi\}\}$
(c) $\{0\}$ (d) $\{0, \phi, \{\phi\}\}$

Ans. (b) : $\{\phi, \{\phi\}\}$

Because If A is a infinite set then set of all subset of A is called power set A denoted by p(A)

Here, A = $\{\phi\}$

P(A) = $\{\phi, \{\phi\}\}$

So answer is (b)

5. If the disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O blocks requests are 98, 37, 14, 124, 65, 67

(a) 239 (b) 310

(c) 321 (d) 325

Ans. (c) : Because according to FCFS algorithm, Total disk moves = (98–32) + (98–37) + (37–14) + (124–14) + (124–65) + (67–65) = 321

So answer is (c)

6. Component level design is concerned with

(a) Flow oriented analysis

(b) Class based analysis

(c) Both of the above

(d) None of the above

Ans. (c) : Because Analysis model, manifested by scenario based, class based, flow oriented and behavioral elements, feed the design task.

using design notations and design methods, design produces a data design, an architectural design, an interface design and a component level design.

7. The 'C' language is

(a) Context free language

(b) Context sensitive language

(c) Regular language

(d) None of the above

Ans. (b) : Most of the programming languages like C, C++, Java etc. can be well approximated by CFG, and the compilers are made taking into account CFGs.

However, C language itself contains context sensitive properties

8. The Mobile Application Protocol (MAP) typically runs on top of which protocol?

(a) SNMP (Simple Network Management Protocol)

(b) SMTP (Simple Mail Transfer Protocol)

(c) SS7 (Signalling System 7)

(d) HTTP (Hyper Text Transfer Protocol)

Ans. (c) : Because, main SS7 functions are

- Setting up and tearing down circuit switched connections
- Advanced network features such as those offered by supplementary service
- SMS & EMS
- Mobility Management in Cellular Networks
- Support for in services

9. If a packet arrive with an M-bit value is 'I' and a fragmentation offset value '0', then it is —— fragment.

(a) First (b) Middle
(c) Last (d) All of the above

Ans. (a) : Because, M = 1 indicates that this packet is not the last packet among all fragments
offset = 0 means this packet carries 0 * 8 = 0^{th} byte from the original packet
Hence answer is (a)

10. The number of bit strings of length eight that will either start with a 1 bit or end with two bits 00 shall be

(a) 32 (b) 64
(c) 128 (d) 160

Ans. (d) : Because string starting with 1-8 places, 1 places fixed, 7 places have 2 choices 2 ^ 7 = 128
ending with 00 - 8 places 2 fixed = 2 ^ 6 = 64
Common string will be there that have been counted twice are starting with 1 and ending with 00. Such number of string will be.
Fix 3 position rest have a choices = 32
total = 128 + 64– 32= 160

11. In compiler design 'reducing the strength' refers to

(a) reducing the range of values of input variables.
(b) code optimization using cheaper machine instructions.
(c) reducing efficiency of program
(d) None of the above

Ans. (b) : Strength reduction is a compiler optimization technique where expensive operations are replaced with equivalent but less expressive operations. The classic example of strength reduction converts "strong" multiplications inside a loop weaker addition. Something that frequently occurs in array addressing.

12. In which addressing mode, the effective address of the operand is generated by adding a constant value to the contents of register?

(a) Absolute (b) Indirect
(c) Immediate (d) Index

Ans. (d) : Index mode: The address of the operand is obtained by adding to the contents of the general register (called index register) a constant value.

13. **Which of the following is true?**
(a) A relation in BCNF is always in 3NF.
(b) A relation in 3NF is always in BCNF.
(c) BCNF and 3NF are same.
(d) A relation in BCNF is not in 3NF.

Ans. (a) : BCNF: FD $\underline{x} \to \overline{y}$, where x is superkey of relation and y is attribute(s).
3NF: FD x→y, where either x is a superkey of relation or y is a prime attributes.
There, BCNF is stronger than 3NF, means 3NF relations are subset of BCNF relations but converse may not be true.

14. **Given memory partitions of 100 K, 500 K, 200 K, 300 K and 600 K (in order) and processes of 212 K, 417 K, 112 K, and 426 K (in order), using the first-fit algorithm, in which partition would the process requiring 426 K be placed?**
(a) 500 K (b) 200 K
(c) 300 K (d) 600 K

Ans. (*) 400 K cannot be placed. So, none option is true.

15. **What is the size of the Unicode character in Windows Operating System?**
(a) 8-Bits (b) 16-Bits
(c) 32-Bits (d) 64-Bits

Ans. (b) : Because, Unicode is a worldwide character encoding standard. The system uses unicode exclusively and string manipulation.
These function use UTF – 16 (wide character) encoding. Which is most common encoding of unicode and the one used for native unicode encoding on windows operating system..

16. **In which tree, for every node the height of its left subtree and right subtree differ almost by one?**
(a) Binary search tree
(b) AVL tree
(c) Threaded Binary Tree
(d) Complete Binary Tree

Ans. (b) : AVL tree is a self balancing binary search tree where the difference between heights of left and right subtrees cannot be more than one for all nodes. If at any time they differ by more than one, nodes. If at any time they differ by more than one, rebalancing is done to restore this property.
Hence the correct answer is (b)

17. **The design issue of Datalink Layer in OSI Reference Model is**
(a) Framing
(b) Representation of bits
(c) Synchronization of bits
(d) Connection control

Ans. (a) : Because, framing is related to data link layer and option (b) and (c) are related to physical layer and option (D) is related to transport layer means framing is done at the data link layer whereas fragmentations is done as network layer and segmentation is done at transport layer.

18. Given the following expressions of a grammar

$$E \rightarrow E * F / F + E / F$$

$$F \rightarrow F - F / id$$

Which of the following is true?

(a) * has higher precedence than +
(b) – has higher precedence than *
(c) + and – have same precedence
(d) + has higher precedence than *

Ans. (b) : Because to check the precedence check the level in which the operator occurs lower the leveled higher the priority and vice versa as – is lower *, – has higher priority than *
So answer is (b)

19. The maturity levels used to measure a proces are

(a) Initial, Repeatable, Defined, Managed, Optimized.
(b) Primary, Secondary, Defined, Managed, Optimized.
(c) Initial, Stating, Defined, Managed, Optimized.
(d) None of the above

Ans. (a) : Because, capability maturity model (CMM) is a development model created after a study of data collected from organizations. The term "maturity" relates to degree of formality and optimization of processes to formality defined steps, to managed result metrics, to active optimization of the process.

CMM Levels

Level 1 > initial
Level 2 > Repeatable
Level 3 > Defined
Level 4 > Managed
Level 5 > Optimized

20. The problem of indefinite blockage of low-priority jobs in general priority scheduling algorithm can be solved using :

(a) Parity bit (b) Aging
(c) Compaction (d) Timer

Ans. (b) : Because a major problem with priority scheduling is indefinite blocking or starvation. A solution to the problem of indefinite blockage to the low priority process is aging. Aging is a technique of gradually increasing the priority of processes that wait in the system for a long period of time.
So option (b) is right choice.

21. Which API is used to draw a circle?

(a) Circle ( ) (b) Ellipse ( )

(c) Round Rect ( ) (d) Pie ( )

Ans. (b) : Because, Ellips is the API which is use to draw a circle.
example : ellipse (56, 46, 55, 55);
⇒ An ellipse with an equal height and width is a circle. The first two parameters set the location, third sets the width and fourth sets the height.

22. In DML, RECONNECT command cannot be used with
(a) OPTIONAL Set (b) FIXED Set
(c) MANDATOR Set (d) All of the above

Ans. (b) : Fixed: a member record cannot exist on its own. Moreover, once it is inserted in a set occurrence, it is fixed if cannot be reconnected to another set occurrence.
Reconnect command: if the connection to the server is lost, automatically try to reconnect attempt is made each time the connection is lost. To suppress reconnection behavior, use "skip-reconnect".

23. Coaxial cables are categorized by Radio Government rating are adapted for specialized functions. Category RG-59 with impedance 75Ω used for
(a) Cable TV (b) Ethernet
(c) Thin Ethernet (d) Thick Ethernet

Ans. (a) : Because, Coaxial cable is a type of electrical cable that has an inner conductor surrounded by a tubular insulating layer and RG – 59/U is a specific type of Coaxial cable, often used for low-power video and RF signal connections. RG–59 Coaxial cable is commonly packed with consumer equipment, such as VCP's or digital cable receivers.

24. RAD stands for
(a) Rapid and design
(b) Rapid Aided Development
(c) Rapid Application Development
(d) Rapid Application Design

Ans. (c): Rapid application development (RAD) is a software development methodology that uses minimal planning in favor of rapid prototyping. A prototype is working model which is functionality equivalent to a component of the product.

25. Suppose that someone starts with a chain letter. Each person who receives the letter is asked to send it on to 4 other people. Some people do this, while some do not send any letter. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? Also find how many people sent out the letter?
(a) 122 & 22 (b) 111 & 11
(c) 133 & 33 (d) 144 & 44

Ans. (c) : Because, from basic, consider one sender, he sent to four other who only read "two" they send to seven other "three" they send to ten other receiver who only recall this form AP having common difference–3, & a/q total number of receiver who only read is 100 that is over last term.

So, 4 + (n–1) * 3 = 100

n = 33 this is the number of reader who send to four other already there are 100 viewer who only read

So total viewer will 100 + 33 = 133

26. A hash function f defined as f (key) = key mod 13, with linear probing is used to insert keys 55, 58, 68, 91, 27, 145. What will be the location of 79?

(a) 1 (b) 2

(c) 3 (d) 4

Ans. (d) : Given, hash function f(key) = key mod 13 with linear probing to resolve collision.

Key are : 55, 58, 68, 71, 27, 145 and 79

55 mod 13 = 3

58 mod 13 = 6

65 mod 13 = 3 (collision, so next) = 4

91 mod 13 = 0

27 mod 13 = 2

145 mod 13 = 2

79 mod 13 = 1 (collision, so next

= 2 (again collision, so next)

= 3 (again collision, so next)

= 4 (again collision, so next)

= 5

So, index 5 is alloted to key 79.

27. Which of the following is true while converting CFG to LL (I) grammar?

(a) Remove left recursion alone

(b) Factoring grammar alone

(c) Both of the above

(d) None of the above

Ans. (c): LL (1) is top-down parser, for top down parser the grammar should be unambiguous, deterministic and free from infinite loop. That means grammar should be free from ambiguity, left factoring and left recursion. All the 3 condition must be satisfied for LL (1) parser.

So option (c) is the right answer.

28. Identify the Risk factors which are associated with Electronic payment system.

(a) Fraudulent use of Credit Cards.

(b) Sending Credit Card details over internet.

(c) Remote Storage of Credit Card details.

(d) All of the above

Ans. (d) : All are given risk factors that which are associated with electronic payment system.

- Fraudulent use of credit card
- Sending credit card details over internet
- Remote storage of credit card details

So answer is (D)

29. Which of the following are two special functions that are meant for handling exception that occurs during exception handling itself?

(a) Void terminate ( ) and Void unexpected ( )
(b) Non void terminate ( ) and void unexpected ( )
(c) Void terminate ( ) and non void unexpected ( )
(d) Non void terminate ( ) and non void unexpected ( )

Ans. (a) : The exception handling mechanism relies on two function, terminate () and unexpected (), for coping with errors related to the exception handling mechanism itself.

Void terminate ()

- If an exception is thrown but not caught
- If the exception handling mechanism find the stack is corrupted.
- If a destructor propagates an exception during stack unwinding due to another exception.

Void unexpected

If a function throws an exception by its exception specification then:

- The stack is unwounded for the function
- The function unexpected () is called

30. Which of the following memory allocation scheme suffers from external fragmentation?

(a) Segmentation
(b) Pure demand paging
(c) Swapping
(d) Paging

Ans. (a): Segmentation is a memory management technique in which each job is divided into several segments of different sizes, one for each module that contains pieces that perform related function. Each segment is loaded into a contiguous block of available memory.

External fragmentation exists when total free memory is enough for the new process but it's not contiguous and cannot satisfy the request. Segmentation suffers from external fragmentation and paging suffers from internal fragmentation.

31. Basis path testing falls under

(a) system testing
(b) white box testing
(c) black box testing
(d) unit testing

Ans. (b): White Box testing is technique that examines the program structure and derives test data from the program code.

Basic path testing or structural testing is white box method for designing test cases. The method analyzes the control flow graph of program to find a set of linearly independent paths of execution.

32. **The User Work Area (UWA) is a set of Program variables declared in the host program to communicate the contents of individual records between**

(a) DBMS & the Host record

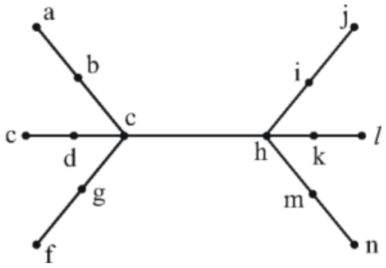
(b) Host program and Host record

(c) Host program and DBMS

(d) Host program and Host language

Ans. (c): The User Work Area (UWA) is set of program variables declared in host program to communicate contents of individual records between DBMS and host program, one program variable for each record type with same format.

33. **Consider the tree given below :**



(a) d & h (b) c & k

(c) g, b, c, h, i, m (d) c & h

Ans. (d) : Eccentricity of vertex : The maximum distance between a vertex to all other vertices is considered as the eccentricity of vertex denoted by (v). The distance from a particular vertex to all other vertices in the graph is taken and among those distances, the eccentricity is the highest of distances for this graph,

e(a) = 5	e(b) = 4	e(c) = 3
e(d) = 4	e(e) = 5	e(f) = 5
e(a) = 4	e(h) = 3	e(i) = 4
e(j) = 5	e(k) = 4	e(l) = 5
e(m) = 4	e(n) = 5	

Radivs

r (c) = 3 & r (n) = 3 which is the minimum eccentricity for c 7 h.

Center

e (v) = r (v),

Then v is the central point of graph G.

Here

e(o) = r (c) = 3 & e(h) = r(h) = 3

Hence both c & h are center of tree

Hence option (d) is right answer.

34. **The maximum number of keys stored in a B-tree of order m and depth d is**

(a) md + 1–1

(b) $\frac{\text{md}+1-1}{\text{m}-1}$

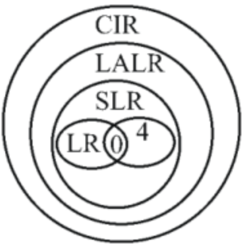
(c) (m – 1) (md + 1 – 1)

(d) $\frac{md-1}{m-1}$

Ans. (a)
$n_o \rightarrow (m-1)$ keys children m, node 1
$n_1 \rightarrow m \times (m-1)$ keys children
$n_2 \rightarrow m \times m \times (m-1)$ keys
.
.
$n_d \rightarrow m^d \times (m-1)$ keys
$= (m-1) + m(m-1) + m^2(m-1)$
$+ \text{-------} \; m^d (m-1)$
$= (m-1) . (1 + m + m^2 \text{.........} m^d)$
$= (m-1)((m^{(d+1)-1}) / m-1)$
$= m^{(d+1)-1}$ should be ans.

35. Which of the following is the most powerful parring method?
(a) LL(1)
(b) Canonical LR
(c) SLR
(d) LALR

Ans. (b) : Canonical LR
Because, LR > LALR > SLR



A canonical LR parser or LR(1) parser is a LR(k) parser for K=1, i.e., with single look ahead terminal.
LR(1) parser are more powerful than LALR parsers and LALR parsers are more powerful than SLR.

36. In UNIX, which of the following command is used to set the task priority?
(a) init
(b) nice
(c) kill
(d) PS

Ans. (b): Nice command directly maps to a kernel call of the same name. Nice is used to invoke a utility or shell script with a particular priority, thus giving the process more or less CPU time than other processes.

37. AES is a round cipher based on the Rijndal Algorithm that uses a 128-bit block of data. AES has three different configurations. —— rounds with a key size of 128 bits, rounds with a key size of 192 bits and rounds with a key size of 256 bits.
(a) 5, 7, 15
(b) 10, 12, 14
(c) 5, 6, 7
(d) 20, 12, 14

Ans. (b) AES is a round cipher based on the Rijndael algorithm that uses a 128 bit block of data.
10 rounds with a key size of 128 bits, 12 rounds with a key size of 192 bits, 14 rounds with a key size of 256 bits.

38. Match the following IC families with their basic circuits:

a. TTL	1. NAND
b. ECL	2. NOR
c. CMOS	3. Inverter

Code :

	a	b	c
(a)	1	2	3
(b)	3	2	1
(c)	2	3	1
(d)	2	1	3

Ans. (a)

(a) TTL–Transistor–transistor logic is a logic family built from bipolar junction transistors. It satisfies the NAND.

(b) ECL–Emitter coupled logic is a high speed integrated circuit bipolar transistor logic family. It satisfies the NOR.

(c) CMOS – Complementary metal oxide semiconductor is a technology for constructing Integrated Circuits. It satisfies inverter.

So answer is (a).

39. Match the following with respect to C++ data types :

a. User defined type	1. Qualifier
b. Built in type	2. Union
c. Derived type	3. Void
d. Long double	4. Pointer

Code :

	a	b	c	d
(a)	2	3	4	1
(b)	3	1	4	2
(c)	4	1	2	3
(d)	3	4	1	2

Ans. (a)

(a) User defined data type–This technique of grouping different values is referred to as class. It offers three techniques of defining a new data type, a structures class, a union, so it satisfies (2) union.

(b) Built in type–This is use to store information of various data types. So it is a void.

(c) Derived type–Derived type can mean a composite data type, a subtype, a derived class, So it satisfies (4) pointer

(d) Long double–It refers to a floating point data type, 8 it satisfies qualifier.

40. Given an empty stack, after performing push (1), push (2), Pop, push (3), push (4), Pop, Pop, push (5), Pop, what is the value of the top of the stack?

(a) 4 (b) 3

(c) 2 (d) 1

Ans. (d) Because operation

Push (1)	Push (2)	Pop	Push (3)	Push (4)	Pop	Pop	Push (5)	Pop
1	2 1	1	3 1	4 3 1	3 2	1	5 1	1

Push is the operation that which adds an element to the collection, pop that is present in the stack. Therefore, popped sequence is: 2, 4, 3and 5 remaining element to be popped is only 1 on the stack. So answer is (d)

41. Enumeration is a process of

(a) Declaring a set of numbers
(b) Sorting a list of strings
(c) Assigning a legal values possible for a variable
(d) Sequencing a list of operators

Ans. (c): An enumeration is a complete ordered listing of the entire item in a collection. The term is commonly used in mathematic and computer science to refer to a listing of all of the elements of a set.

Only certain pre-defined values are allowed.

Each valid value is assigned a name, which is then normally used instead of integer when working with this data type.

42. Which of the following mode declaration is used in C++ to open a file for input?

(a) ios : : app
(b) in : : ios
(c) ios : : file
(d) ios : : in

Ans. (d) : iOS :: in//open for input operations.

43. Data Encryption Techniques are particularly used for ———.

(a) protecting data in Data Communication System.
(b) reduce Storage Space Requirement.
(c) enhances Data Integrity.
(d) decreases Data Integrity.

Ans. (a) : Because, Data encryption is a security in which information is encoded in such away that only authorized reader can read it and data encryption is particularly used for protecting data in data communication system.

So answer is (a).

44. Let L be a set accepted by a non-deterministic finite automaton. The number of states in non-deterministic finite automaton is $|Q|$. The maximum number of states in equivalent finite automaton that accepts L is

(a) $|Q|$ (b) $2|Q|$

(c) $2^{|Q|}-1$ (d) $2^{|Q|}$

Ans. (d): Because, conversion from NFA to DFA is done by subset construction. It a problem can be solved with state in NFA then in worst case number of states in the resulting DFA is 2^n

Given number of states in NFA = $|Q|$

Then maximum number of states in equivalent

DFA = $2^{|Q|}$

Hence, option (d) is correct choice.

45. What is the result of the following expression?

(1 & 2) + (3 & 4)

(a) 1 (b) 3

(c) 4 (d) 0

Ans. (d): Given (1&2) + (3&4) in decimal

= (001 & 010) + (011 & 100) in binary

= (000) + (000) = (000)

=0 in decimal.

Hence answer is (d).

46. Back propagation is a learning technique that adjusts weights in the neural network by propagating weight changes.

(a) Forward from source to sink

(b) Backward from sink to source

(c) Forward from source to hidden nodes

(d) Backward from since to hidden nodes

Ans. (b): Back propagation, an abbreviation for "backward propagation of error" is a learning technique that adjusts weights in neural network by propagating weight changes backward from the sink to the source nodes.

47. Match the following:

a. TTL	1. High fan out
b. ECL	2. Low propagation delay
c. CMOS	3. High power dissipation

Code:

	a	**b**	**c**
(a)	3	2	1
(b)	1	2	3
(c)	1	3	2
(d)	3	1	2

Ans. (a)

(a) TTL–Transistor-transistor logic is a digital logic design. TTL is characterized by high switching speed. It has high power dissipation.

(b) ECL–Emitter coupled logic is a high speed integrated circuit bipolar transistor logic family. In ECL, transistor are never in saturation, the input/output voltage have a swing (0.8v), the input impedance is high and output resistance is low. So it has low propagation delay.

(c) CMOS – Complementary metal oxide semiconductor is a technology for constructing IC's CMOS technology is used in microprocessors, static RAM & other digital logic circuit. It has high fan out.

48. ——— **is an "umbrella" activity that is applied throughout the software engineering process.**
(a) Debugging
(b) Testing
(c) Designing
(d) Software quality assurance

Ans. (d): Because, umbrella activities include:
(1) Software Project Management
(2) Formal Technical reviews
(3) Software Quality assurance
(4) Measurement
(5) Risk Management
(6) Reducibility Management
Hence (d) is the answer.

49. Identify the operation which is commutative but not associative?
(a) OR (b) NOR
(c) EX-OR (d) NAND

Ans. (d) : Because, O $NAND_1$ = 1 $NAND_0$ = 1 $\Rightarrow$ Cumulative
(O $NAND_1$) $NAND_1$ = 1 $NAND_1$ = 0
O NAND (1$NAND_1$) = 0 $NAND_0$ = 1 $\Rightarrow$ Not associative
- Every logic gate follows commutative law
- AND, OR, Ex–OR, Ex–NOR follows associative law, NOR does not follow associative law.
- AND, OR, follows distributive law. Ex-OR, Ex-NOR, NAND, NOR does not follow distributive law.

So answer is (d).

50. Given a Relation POSITION (Posting No, Skill), then query to retrieve all distinct pairs of posting-nos. requiring skill is
(a) Select p.posting-No, p.posting-No
from position p
where p.skill = p.skill
and p.posting-No < p.posting-No
(b) Select p_1.posting-No, p_2.posting-No
from position p_1, position p_2
where p_1.skill = p_2.skill
(c) Select p_1.posting-No, p_2.posting-No
from position p_1, position p_2
where p_1.skill = p_2.skill
and p_1.posting-No < p_2.posting-No
(d) Select p_1.posting-No, p_2.posting-No
from position p_1, position p_2
where p_1.skill = p_2.skill
and p_1.posting-No = p_2.posting-No

Ans. (c) : Because, we need to join position with itself based on skill.
Then it is given distinct pairs so we need to consider only P_1 Posting – no < p_2. Posting – no.